M&AE 5070 - Flight Dynamics

Chapter 3 - Moment Equation

Companion Notes for Case Study 3

This chapter and the accompanying case study take a closer look at the moment equation **M** , as it plays a fundamental role in many aspects of flight dynamics. In the previous chapter, we saw how the angle of attack had an effect on the velocity and the slope γ. The angle of attack is also a very important safety parameter with respect to stall. We will see how **M** also provides a simple stability criterion for the only longitudinal mode that can become unstable. The objective here is to show how it is possible to balance the pitching moment, that is, how we can obtain $C_m = 0$, and explore the role of longitudinal control on this equilibrium. This will be a preliminary look at actual flight dynamics.

1 Aircraft moment coefficient and static stability

Pitching moment is defined relative to the center of gravity of the aircraft, G. Both lift and drag could in theory create a pitching moment, so C_m is *a priori* a function of both C_z and C_x, so a function of C_z and C_z^2. Non-linearity is an extra complexity we'd love to avoid if possible, let's introduce 2 hypotheses to simplify things a bit:

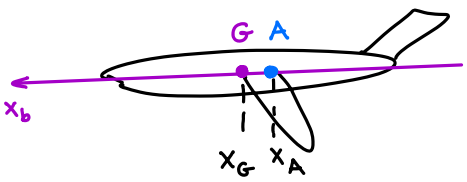
- The C_m coming from C_x is much smaller than that generated by C_z. Either the overall drag force goes through the center of gravity G, or drag is just much smaller than li
- The angle of attack stays far from stall. C_z is linear in $\bar{\alpha}$

C_m linear in C_z (no C_z^2 term) } C_m linear in $\bar{\alpha}$

In this case, experience shows that C_m varies linearly when changing the angle of attack α. That is the basis of our C_m model used in Case Study 2.

Aerodynamic center. The aerodynamic center for an aircraft (also called neutral point) is central to any static and dynamic analysis. The easiest way to define it is the point on the aircraft where a change of lift due to a change of angle of attack will be applied. The change in moment **with respect to the center of gravity** G is equal to the product of the distance between the aerodynamic center A and G, with the change in lift:

R_b frame of reference



$$\Delta M = (x_A - x_G)\,\Delta L$$

* non dimensionalize, $S \equiv$ ref surface area, $l \equiv$ chord
* Look at the change of M and L from a change in α
  L$_b$ divide by $\Delta\alpha$: $\frac{\Delta \cdot}{\Delta \alpha}$
* small changes : $\Delta \to d$

$$\frac{\partial C_m}{\partial \alpha} = \left(\frac{x_A - x_G}{l}\right)\frac{\partial C_z}{\partial \alpha}$$

Eq 1

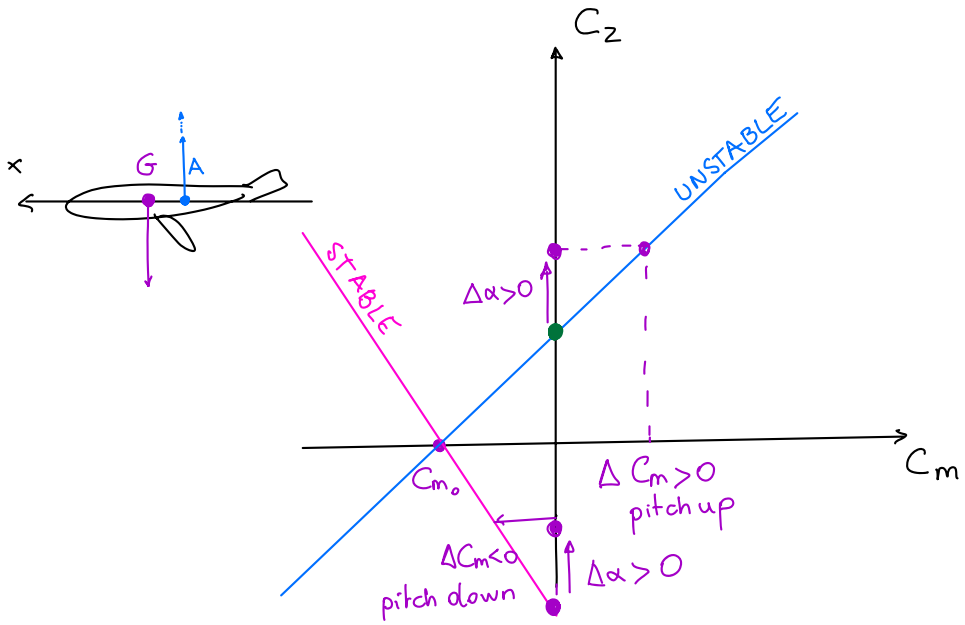
STATIC MARGIN m_s

(Note: m_s has a sign : $x_A - x_G$ is a signed distance along the $\vec{x_b}$ axis)

Integrating the previous equation, we get the corresponding relationship for the moment coefficient:

$$C_m = C_{m_0} + C_{m_\alpha}(\underbrace{\alpha - \alpha_0}_{\bar{\alpha}})$$

Note that at this stage, the aircraft is not trimmed, that is, there is no guarantee that $C_m = 0$. To enable equilibrium flight, we will need to add a contribution from the control δ_m. Let's conveniently put that aside for a moment...



Vertical gust

$\Downarrow$

$\Delta\alpha > 0$

to be statically stable, want the extra created by $\Delta\alpha>0$ to be applies behind the center of gravity

if that is the case,

$\Delta C_m < 0$

$\Downarrow$

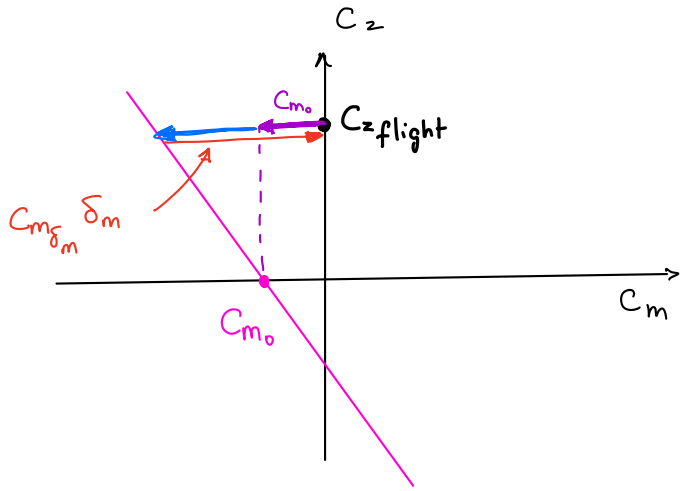
α decreases

$\Rightarrow$ back to initial conditions aircraft stable

For an aircraft to be statically stable, $\frac{\partial C_m}{\partial \alpha} = C_{m_\alpha} < 0$

Static stability : CENTER OF GRAVITY G IN FRONT OF A !

Trimming a stable plane The goal is to ensure that **M** is satisfied: actual flight $C_z > 0$!



$$C_m = C_{m_0} + C_{m_\alpha}(\alpha - \alpha_0) + C_{m_\delta}\delta_m$$

use elevators to balance C_{m_0} AND contributions from lift / α !

- When $C_m = 0$ for cruise conditions, aircraft is said to be **trimmed** : $\delta_m = \delta_{m,trim}$
- In general, $\delta_m = \delta_{m,trim} + \Delta\delta_m$

pilot action on stick during flight

2 Location of aircraft aerodynamic center (neutral point)

Clearly, the location of x_A is quite important for the stability of the aircraft. We want to figure out where it is for a conventional configuration (main wing and tail). Fir this, we first need to clarify a few definitions.

2.1 Revisiting angle definitions

* Angles of attack for lifting surfaces are the angles between the aerodynamic velocity vector [projected onto the vertical plane of symmetry of the aircraft] and the chord line of the cross-section (airfoil) of the lifting surface

need to add this in case of non purely longitudinal flight to get well defined angles

↳ Extension: we define the AIRCRAFT angle of attack α as the angle between [the projection onto the vertical plane of symmetry of the aircraft of] the velocity vector $\vec{V}$ and the fuselage axis $\vec{x_b}$

* The flow of air is deflected by the main wing by an angle ε, so the velocity vector seen by the empennage is **NOT** the main velocity vector

Assuming horizontal empennage, at the rear of the aircraft, potential flow theory provides the following expression:

$$\varepsilon = \frac{2 C_{z_w}}{\pi AR_w}$$

(Prandtl lifting line, deflection is linked to induced drag and the generation of circulation on main wing)

Note: the wing and empennage may be fixed at a non-zero angle on the fuselage, which we will refer to as **offset angles** δ_w and δ_E

δ_w: main wing; δ_E: empennage

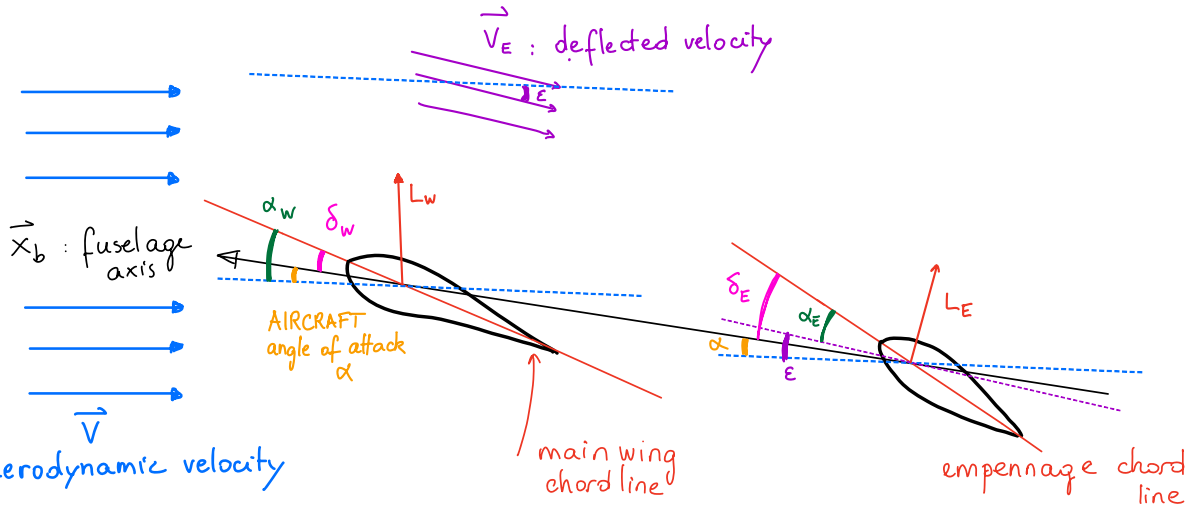
Relationships between angles of attack:

$$\alpha_w = \alpha + \delta_w$$

α_w: angle of attack for main wing; α: Aircraft angle of attack; δ_w: main wing offset [CONSTANT]

$$\alpha_E = \alpha - \varepsilon + \delta_E$$

α_E: angle of attack for empennage; ε: deflection of flow from main wing; δ_E: empennage deflection [CONSTANT]



1.1 Lifting surface-centric (local) vs. aircraft-centric lift coefficients

Lift coefficients are non-dimensional forces, their values therefore depend on what is used to non-dimensionalize the forces. Focusing on the surface area contribution, we can express the lift force in 2 equivalent, non-dimensional form:

$$L_W = \frac{1}{2}\rho S_W V^2 C_{z_W} = \frac{1}{2}\rho S V^2 C^*_{z_W}$$

$$L_E = \frac{1}{2}\rho S_E V^2 C_{z_E} = \frac{1}{2}\rho S V^2 C^*_{z_E}$$

$$\begin{cases} C^*_{z_W} = C_{z_W}\frac{S_W}{S} \\ C^*_{z_E} = C_{z_E}\frac{S_E}{S} \end{cases}$$

↑ same surface area

Can use potential flow theory to evaluate lift coefficients

$$\begin{cases} C_{z_W} = C_{z_{\alpha_W}}(\alpha_W - \alpha_{0_W}) = \dfrac{2\pi}{1+\frac{2}{AR_W}}(\alpha_W - \alpha_{0_W}) \\ C_{z_E} = C_{z_{\alpha_E}}(\alpha_E - \alpha_{0_E}) = \dfrac{2\pi}{1+\frac{2}{AR_E}}(\alpha_E - \alpha_{0_E}) \end{cases}$$

[EDITED 2/24 to correct expression for $C_{z_{\alpha_i}}$]

Notation:

- no double subscript ⇒ aircraft coefficient (C_z)
- double subscript, no asterix: coefficients defined according to potential flow theory, using characteristic quantities associated with the relevant lifting surface.
  C_{z_W}, C_{z_E}, $C_{z_{\alpha_W}}$, $C_{z_{\alpha_E}}$
- double subscript, with asterix: coefficients defined using the aircraft characteristic quantities, accounting for surface areas and flow deflection by the main wing
  $C^*_{z_W}$, $C^*_{z_E}$, $C^*_{z_{\alpha_W}}$, $C^*_{z_{\alpha_E}}$

Forces are additive: $L_{total} = L_W + L_E$, coefficients are additive ONLY if they use the same normalization

↳ $C_z = C^*_{z_W} + C^*_{z_E}$ (but ~~$C_z \neq C_{z_W} + C_{z_E}$~~)

How do lifting surface-centric coefficients differ from their aircraft-centric counterparts?

WING:

REF = AIRCRAFT REF = WING

$$L_w = \frac{1}{2}\rho V^2 S\, C^*_{z_{\alpha,w}}\, \bar{\alpha} = \frac{1}{2}\rho V^2 S_w C_{z_{\alpha,w}}\, \bar{\alpha}_w$$

aircraft wing

α = angle between of aero velocity and **fuselage axis**

α_w = angle between aero velocity and **wing chord line**

and $\alpha = \alpha_w + \delta_w$ ← wing offset

So how do we go from $C^*_{z_{\alpha,w}}$ (aircraft reference) to $C_{z_{\alpha,w}}$ (wing reference)?

$$C_{z_{\alpha,w}} = \frac{\partial}{\partial \alpha_w}(C_{z_w}) = \frac{\partial}{\partial \alpha}(C_{z_w})\frac{\partial \alpha}{\partial \alpha_w} = \frac{\partial}{\partial \alpha}\left(\frac{S}{S_w} C^*_{z_w}\right)\frac{\partial \alpha}{\partial \alpha_w} = \frac{S}{S_w} C^*_{z_{\alpha,w}} \frac{\partial \alpha}{\partial \alpha_w}$$

by definition; change in variable; use renormalization; $= \frac{\partial}{\partial \alpha}(C^*_{z_w})$

So $C^*_{z_{\alpha,w}} = \frac{S_w}{S} C_{z_{\alpha,w}} \underbrace{\frac{\partial \alpha_w}{\partial \alpha}}$

$= \frac{\partial(\alpha + \delta_w)}{\partial \alpha} = 1$ since δ_w is constant

And $\boxed{C^*_{z_{\alpha,w}} = \frac{S_w}{S} C_{z_{\alpha,w}}}$

Ref: S, α Ref: S_w, α_w

EMPENNAGE:

The process is the same, but now $\alpha_E = \alpha + \delta_E - \varepsilon$ ← deflection

$$C_{z_{\alpha,E}} = \frac{\partial}{\partial \alpha_E}(C_{z_E}) = \frac{\partial}{\partial \alpha}(C_{z_E})\frac{\partial \alpha}{\partial \alpha_E} = \frac{\partial}{\partial \alpha}\left(\frac{S}{S_E} C^*_{z_E}\right)\frac{\partial \alpha}{\partial \alpha_E} = \frac{S}{S_E} C^*_{z_{\alpha,E}} \frac{\partial \alpha}{\partial \alpha_E} = C_{z_{\alpha,E}}$$

by definition

So $C^*_{z_{\alpha,E}} = \frac{S_E}{S} C_{z_{\alpha,E}} \underbrace{\frac{\partial \alpha_E}{\partial \alpha}}$

$= \frac{\partial(\alpha + \delta_E - \varepsilon)}{\partial \alpha} = 1 + 0 - \frac{\partial \varepsilon}{\partial \alpha}$

extra term, needs closed-form expression for it.

We have an expression for ε: $\varepsilon = \frac{2 C_{z_w}}{\pi AR_w}$ (deflection depends on main wing: C_{z_w} uses S_w, α_w)

So $\frac{\partial \varepsilon}{\partial \alpha} = \frac{d\varepsilon}{\partial \alpha} = \dot{\varepsilon} = \frac{\partial \varepsilon}{\partial \alpha_w} \frac{\partial \alpha_w}{\partial \alpha}$ ($\frac{\partial \alpha_w}{\partial \alpha} = 1$) $= \frac{2}{\pi AR_w} C_{z_{\alpha,w}}$

↑ from thin airfoil theory $= \frac{2\pi}{1 + \frac{2}{AR_w}}$

$= \frac{2}{\pi AR_w} \frac{2\pi AR_w}{AR_w + 2} = \frac{4}{AR_w + 2}$

$\Rightarrow C^*_{z_{\alpha,E}} = \frac{S_E}{S} C_{z_{\alpha,E}} (1 - \dot{\varepsilon}) = \boxed{\frac{S_E}{S} \frac{AR_w - 2}{AR_w + 2} C_{z_{\alpha,E}} = C^*_{z_{\alpha,E}}}$

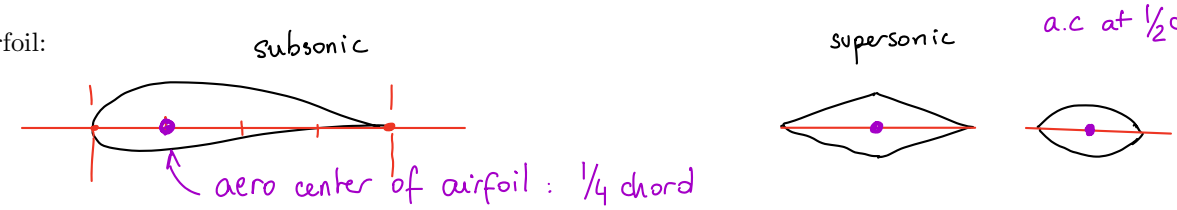
$L_E = \frac{1}{2} \rho S_E C_{z_{\alpha,E}} (\alpha_E - \alpha_{E,zl})$ ← empennage basis

$= \frac{1}{2} \rho S C^*_{z_{\alpha,E}} (\alpha - \alpha_{zl})$ ← aircraft basis

2.3 Aerodynamic center of individual lifting surfaces

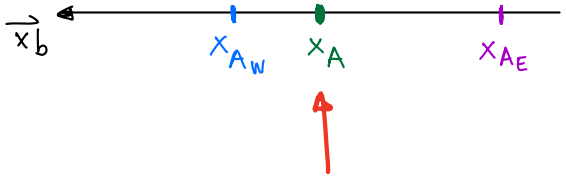
To determine where the aircraft's aerodynamic center is, the first step is to find the aerodynamic center of all lifting surfaces, in our case, **wing** and **empennage** mostly (but also fuselage and nacelles, which we will neglect in our analysis).

- 2D airfoil:



- Finite-length wing: use ¼ chord point of mean chord along wing.

2.4 How is the aircraft aerodynamic center related to the individual ones



$$\boxed{\frac{x_A}{\ell} C^*_{z_\alpha} = \sum_i \frac{x_{A_i}}{\ell} C^*_{z_{\alpha_i}}}$$

↳ "definition" of the location of x_A

Proof. We start with the following equations, all expressed for the aircraft (and keeping in mind that the moment coefficient is linear in C_z)

$$\begin{cases} C_{m_\alpha}^* = \dfrac{x_A - x_G}{\ell} C_{z_\alpha}^* & \text{Eq. 1} \\ C_{m_\alpha}^* = \sum_i C_{m_{\alpha_i}}^* & \text{Eq. 2} \\ C_{z_\alpha}^* = \sum_i C_{z_{\alpha_i}}^* & \text{Eq. 3} \end{cases}$$

* Each component contributing to lift

$$C_{m_{\alpha_i}} = \frac{x_A - x_G}{\ell} C_{z_{\alpha_i}}^* \quad (i \equiv \text{individual component})$$

Eq 2, replacing C_m by C_z $\quad C_{m_\alpha} = \sum_i C_{m_{\alpha_i}}$

$\boxed{1} = \boxed{2}$ because of Eq 3

Expand: $\dfrac{x_A - x_G}{\ell} C_{z_\alpha}^* = \sum_i \dfrac{x_{F_i} - x_G}{\ell} C_{z_{\alpha_i}}^*$

$$\frac{x_A}{\ell} C_{z_\alpha}^* \boxed{- \frac{x_G}{\ell} C_{z_\alpha}^*}^{1} = \sum_i \frac{x_{A_i}}{\ell} C_{z_{\alpha_i}}^* \boxed{- \sum_i \frac{x_G}{\ell} C_{z_{\alpha_i}}^*}^{2}$$

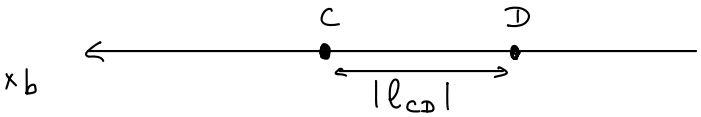
Note: this formula is independent of choice of origin on the $\vec{x_b}$ axis

$$\boxed{\frac{x_A}{\ell} C_{z_\alpha}^* = \sum_i \frac{x_{A_i}}{\ell} C_{z_{\alpha_i}}^*}$$

Let's reformulate that formula to make it a bit more practical. Since the location of the zero on the x_b axis is arbitrary, we can simplify the algebra by placing it at the location of the aerodynamic center. Making the contributions from the lifting surfaces (wings and empennage) explicit: $x_A = 0$

$$\frac{x_{A_W}}{\ell} C_{z_{\alpha,W}}^* + \frac{x_{A_E}}{\ell} C_{z_{\alpha,E}}^* = 0$$

Let's introduce a short-hand notation for the various distances involved:

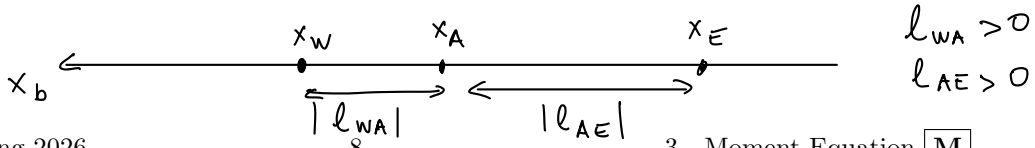


$$\ell_{CD} = x_C - x_D$$

Remember that all length except for the reference length ℓ have a sign, the order of the subscript does indeed matter.

- Reference length: Wing mean aerodynamic chord, always positive: ℓ
- Distance between aerodynamic centers Wing - Aircraft: $\ell_{WA} = x_W - x_A$
- Distance between aerodynamic centers Empennage - Aircraft: $\ell_{EA} = x_E - x_A$
- Distance between wing's aerodynamic center and center of gravity: $\ell_{WG} = x_W - x_G$
- Distance between empennage's aerodynamic center and center of gravity: $\ell_{EG} = x_E - x_G$
- Distance between aerodynamic centers Wing - Empennage (stagger): $d = \ell_{WA} - \ell_{EA} = \ell_{WG} - \ell_{EG} = \ell_{WE} > 0$
- Static margin (*i.e.* distance between center of gravity and aircraft aerodynamic center):

$$m_s = \frac{x_A - x_G}{\ell} = -\frac{\ell_{GA}}{\ell} \tag{1}$$



Plugging those results into the definition of x_A provides a more convenient expression for the ratio $\frac{\ell_{WA}}{\ell_{EA}}$:

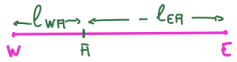
$$\ell_{WA}\, C^{*}_{z_{\alpha,W}} + \ell_{EA}\, C^{*}_{z_{\alpha,E}} = 0$$

$$\ell_{WA}\, S_W\, C_{z_{\alpha,W}} + \ell_{EA}\, S_E \left(\frac{AR_W - 2}{AR_W + 2}\right) C_{z_{\alpha,E}} = 0$$

↓ Switch from aircraft to component reference

$$\Rightarrow \quad \frac{x_W - x_A}{x_E - x_A} = \frac{\ell_{WA}}{\ell_{EA}} = -\frac{S_E}{S_W}\left(\frac{AR_W - 2}{AR_W + 2}\right)\frac{C_{z_{\alpha,E}}}{C_{z_{\alpha,W}}}$$

↑ from potential flow theory, function of AR_W and AR_E only

a bit of algebra later... 

$$\boxed{\frac{\ell_{WA}}{\ell_{EA}} = -\frac{S_E}{S_W}\frac{1 - 2/AR_W}{1 + 2/AR_E}} = -R$$

→ position of x_A is only a function of geometry of aircraft.

↖ for later convenience

2.5 Ratio of lift coefficients between wing and empennage

Let's now try to figure out the distribution of lift forces between wing and empennage, for example by evaluating the ratio

$$\frac{C_{z_W}}{C_{z_E}} \tag{2}$$

Note: no *: those are using the local components surface area in the normalization of the lift force.

Moment balance on the aircraft. With respect to the center of gravity of the aircraft, G:

$$M = 0 = M_0 + L_W\, \ell_{WG} + L_E\, \ell_{EG}, \quad \text{with} \quad M'_0 = \frac{1}{2}\rho S V^2 \ell\, C_{m_0}$$

↖ mean aero chord

We can replace L_W and L_E by their expression:

$$\boxed{S_W\, C_{z_W}\, \ell_{WG} + S_E\, C_{z_E}\, \ell_{EG} = -S\ell\, C_{m_0}}$$

↖ NO star

We can also express the total lift force on the aircraft:

$$mg = L_W + L_E = \frac{1}{2}\rho V^2 S C_z$$

(S: aircraft ref surface; C_z: aircraft C_z)

assume ε small so that $|\vec{V}| \simeq |\vec{V_E}|$

$$\Rightarrow \quad \boxed{S_W C_{z_W} + S_E C_{z_E} = S C_z}$$

Taking the ratio of those last 2 expressions, and defining $\ell_0 = -\ell \frac{C_{m_0}}{C_z}$:

$$\frac{S_W C_{z_W} \ell_{WG} + S_E C_{z_E} \ell_{EG}}{S_W C_{z_W} + S_E C_{z_E}} = \ell_0$$

$$\Rightarrow \quad S_W C_{z_W} \ell_{WG} + S_E C_{z_E} \ell_{EG} = S_W C_{z_W} \ell_0 + S_E C_{z_E} \ell_0$$

$$\Rightarrow \quad C_{z_W} S_W (\ell_{WG} - \ell_0) = - C_{z_E} S_E (\ell_{EG} - \ell_0)$$

$$\Rightarrow \quad \boxed{\frac{C_{z_W}}{C_{z_E}} = - \frac{S_E}{S_W} \frac{\ell_{EG} - \ell_0}{\ell_{WG} - \ell_0}}$$

To relate that to the expression we obtained for the aerodynamic center, let's play around with lengths:

* $\ell_{GA} = - m_s \ell$ by definition
* $\ell_{WG} + \ell_{GE} = d$ (distance wing-empennage)
* $\ell_{WA} = - R \ell_{EA}$ (from previous page, def of aero center)

using

$$\Rightarrow \ell_{WG} + \ell_{GA} = - R(\ell_{EG} + \ell_{GA})$$

$$\Rightarrow (d + \ell_{EG}) + \ell_{GA} = - R \ell_{EG} - R \ell_{GA}$$

$$\Rightarrow \quad \ell_{EG}(1+R) = -d - (1+R)\ell_{GA}$$

$$\boxed{\ell_{EG} = - \frac{d}{1+R} - \ell_{GA}}$$

In a similar way, we can show that

$$\boxed{\ell_{WG} = \frac{R}{1+R} d - \ell_{GA}}$$

Finally, bringing everything back together (skipping a few algebra steps), we obtain

$$\boxed{\frac{C_{z_W}}{C_{z_E}} = \frac{S_E}{S_W} \frac{1 + (R+1)B}{R - (R+1)B}} \quad (4)$$

$\frac{C_{z_W}}{C_{z_E}}$ is a function of geometry of aircraft, static margin AND flight conditions through C_z

with

$$\boxed{B = - \frac{\ell}{d} \left(m_s + \frac{C_{m_0}}{C_z} \right)} \quad (5)$$

Why do we care about those expressions: they express the lift coefficient ratio as a function of geometric parameters, and 2 important design and stability parameters: the overall aircraft lift coefficient, and static margin.